

**Advancing drug discovery
through rapid, high-resolution
visualization of organoids and
spheroids with LICORbio
ATLAS™ IMAGER**

**High-content imaging of 3D cell models with the LICORbio
Atlas™ Imager**

Biomedical research is undergoing a significant shift towards more physiologically relevant models for studying human biology and disease. Traditional two-dimensional (2D) cell cultures, while useful, often fail to replicate the complex structure and functionality of human tissues. These limitations, as well as the FDA's recently announced plan to phase out animal testing requirements for drug development in favour of New Approach Methodologies (NAMs), have spurred the development and reinforced the adoption of three-dimensional (3D) cell culture systems such as organoids and spheroids.

These 3D cell culture models generally provide a much closer approximation of true human in vivo conditions than 2D cell culture and animal models, making them invaluable tools for drug development, disease modelling, and personalized medicine.

This application provides a basic overview of organoids and spheroids, as well as their similarities, differences, applications, and implications for the future of drug discovery. It also highlights technologies like the LICORbio ATLAS™ Imager that bridge the gap between complexity and clarity—delivering high-content, high-resolution imaging of 3D models.

Keywords or phrases:

3D cell culture imaging, Organoids and spheroids, High-content, high-throughput imaging, Multiplex fluorescence assays, Drug discovery and disease modelling, Spatial biology and quantitative analysis

INTRODUCTION

What Are Organoids?

Organoids are miniature, self-organizing 3D structures derived from stem cells or primary tissues that can simulate the structure and function of real organs. They are typically

cultured in extracellular matrix scaffolds (often hydrogels like Matrigel or synthetic alternatives), which provide the necessary support for cell growth and differentiation. Organoids can be generated from various tissues, including the brain, liver, lung, retina, intestine, and kidney, and they contain multiple cell types found in the corresponding organ. This cellular diversity allows organoids to closely mimic the physiological environment of the original tissue.

The practical applications of organoids are incredibly diverse. They are used in drug screening to assess efficacy and toxicity, in disease modelling to study pathogenesis, and in regenerative medicine to explore tissue development and repair mechanisms. Furthermore, organoids have been vital to personalized medicine by enabling the testing of patient-specific therapies to identify the most effective treatment options.

What Are Spheroids?

Spheroids are free-floating, 3D cell aggregates formed from cell suspensions that spontaneously assemble into spherical structures, with or without an extracellular scaffold matrix. Unlike organoids, spheroids are simpler in structure and often only consist of a single cell type or limited number of cell types. They are typically cultured in non-adherent conditions that promote cell aggregation, such as ultra-low attachment (ULA) plates, hanging drop systems, liquid overlay, spinner or pellet culture, rotating wall vessels, microfluidics, or via magnetic levitation.

Despite their relative simplicity, spheroids are valuable models for studying cellular behaviours like proliferation, migration, and drug response via cell health assays. This makes them particularly useful in drug development, as well as cancer research, where they serve as models for tumour growth and metastasis. Spheroids can also be utilized in high-throughput screening assays due to their ease of formation and scalability.

Organoids vs. Spheroids

Organoids and spheroids are both 3D cell culture models and offer more physiologically relevant alternatives than traditional 2D cell cultures. However, they differ

significantly in their formation, complexity, and applications. While both models have their strengths, the choice between organoids and spheroids depends on the specific research objectives and the level of complexity required.

1. Complexity and Structure

Organoids and spheroids are quite different in terms of relative complexity. Organoids are far more complex, as they contain multiple cell types and exhibit tissue-specific architecture. Spheroids are simpler, often composed of a single cell type, and lack the intricate structures seen in organoids.

Organoids exhibit a high degree of complexity, often developing organized structures that resemble the original tissue. This complexity arises from the self-organizing properties of stem cells, which can differentiate into various lineages and form functional biological tissues. For example, intestinal organoids can develop villus-like structures and exhibit absorptive and secretory functions like the human intestine. Organoids provide a more detailed and accurate representation of tissue architecture, making them suitable for studies requiring complex models, such as disease modelling and personalized medicine.

Spheroids are much simpler in structure. They typically consist of a mass of cells without the distinct layers or compartments seen in organoids. While they can form from various cell types and may exhibit some degree of cellular heterogeneity, spheroids inherently lack the intricate organization and functionality of organoids, limiting their ability to fully mimic the functionality of true human tissues. While spheroid models may be considered reductionist compared to organoids, their ease of generation makes them more practical for high-throughput protocols and drug screening.

2. Cellular Composition

Organoids are composed of multiple cell types that are spatially organized to mimic the original tissue. This multi-

cellular composition arises from the differentiation of stem or progenitor cells under specific culture conditions. The presence of various cell types within organoids allows for more accurate modelling of tissue-specific functions and interactions. For instance, brain organoids can contain neurons, glial cells, and other supporting cells, providing a more comprehensive model of neural development and disease.

Spheroids, in contrast, often consist of a single cell type or a limited number of cell types. This simplicity can be quite advantageous for certain applications, such as studying basic cellular behaviours or conducting high-throughput drug screening despite not fully capturing the complexity of true human tissues.

3. Applications

Both organoids and spheroids can be found in various biomedical research applications. The choice to use organoids or spheroids generally depends on the specific research question and the level of complexity required. In some cases, combining both models can provide complementary insights into biological processes and therapeutic responses. In general, organoids are suited for long-term culture, genetic manipulation, and modelling organ-specific diseases. Spheroids are advantageous for cell health assays, rapid drug screening, and studying basic cellular behaviours.

Spheroids are widely utilized in cancer research due to their ability to mimic aspects of tumour growth and the tumour microenvironment. They are particularly useful for evaluating drug efficacy and resistance, as they can replicate the 3D architecture of tumours and the gradients of nutrients and oxygen found within them. Additionally, spheroids have the potential to be more common in high-throughput screening, making them valuable tools for drug discovery.

Organoids, however, offer a more advanced model for studying human biology and disease. Their complex structure and cellular composition make them ideal for modelling organ-specific diseases, such as cystic fibrosis in lung organoids or colorectal cancer in intestinal organoids.

Organoids also facilitate personalized medicine approaches, as patient-derived organoids can be used to test drug responses and tailor individual treatments.

While 2D cell culture is still the most popular approach, spheroids can provide value for applications requiring simplicity and rapid scalability. Organoids offer a more detailed and physiologically relevant model for studying complex biological systems and diseases, though they tend to be relatively costly and time-consuming to produce and maintain.

HIGH-CONTENT, HIGH-THROUGHPUT IMAGING WITH THE LICORBIO ATLAS IMAGER

Designed to streamline imaging workflows, the ATLAS integrates advanced optics and automation to support multiplexed assays with minimal user intervention. A unified solution of high-throughput and deep spatial resolution, the system reduces workflow complexity, minimises steps and accelerates image acquisition—delivering results in a fraction of the time and cost compared to conventional imaging platforms.

KEY FEATURES

- ▶ Acquire entire wells in an image to perform rapid, full-plate screening, and conduct higher-resolution deep dives (5 μm) into individual wells and regions of interest.

No post-processing manipulation needed, such as

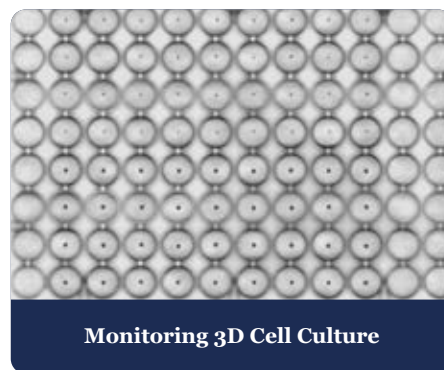
- ▶ stitching or illumination correction.

The unique line-scanning technology enables whole-plate imaging in less than one minute.

- ▶ The patented optical system gives you unmatched sensitivity and dynamic range in a single scan. It minimizes background fluorescence and enhances quantitative multiplexing for critical drug discovery applications.

- ▶ Over 30 imaging channels with six lasers spanning UV to near-infrared, plus LED RGB for brightfield and epi-illumination.
- ▶ Perform cell viability and luminescence assays with the built-in luminescent imager.
- ▶ The expansive focus depth enables imaging through entire 3D cell structures such as spheroids, organoids, organ-on-a-chip, and other micro physiological systems.

Shown below are example images acquired using the LICORbio ATLAS Imager, illustrating its ability to capture both 2D and 3D assay data with exceptional clarity and contrast.



Monitoring 3D Cell Culture

PANC-1 and BxPC3 spheroids grown in a 96-well plate, imaged in 630 nm trans channel at 50 μm resolution.



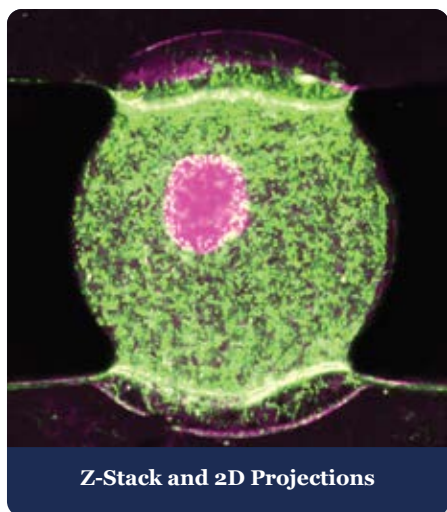
Single Well Imaging in Brightfield

Panc-1 spheroid grown in a 96-well plate imaged in the 630 trans channel at 5 μm .



Monitoring 3D Cell Culture

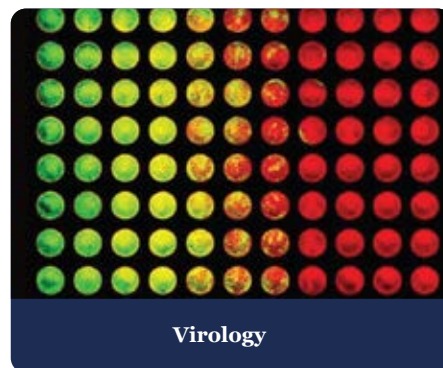
Brightfield image (RGB transillumination) of spheroid in single well image acquired at 5 μm on the LICORbio Atlas Imager. RGB transillumination enables crisp, color detail of the contents of the entire well.



Z-Stack and 2D Projections

Multiplexed fluorescence imaging of a vascularized, liver cancer spheroid in an organiX Plate from AIM Biotech. The sample was imaged using the LICORbio Atlas Imaging System at 5 μm resolution, capturing the entire well in a single image, and Image Studio Software v6.1 was used to create a max projection. The image shows Hoechst (DNA stain) in the 375Ex-460Em channel (blue), GFP expressed in the vasculature in the 488Ex-530Em channel (green), and

Phalloidin555 bound to F-actin in the 520Ex-590Em channel (red). Co-localization is displayed for Hoechst/Phalloidin555 (magenta) and GFP/Phalloidin555 (yellow).



Virology

Easily perform fast, reliable, and quantitative virology assays using near-infrared fluorescence imaging.



Tissue Section Analysis

Tissue section at 5 μm shown with automatically generated shape contour using new Morphology Analysis features in Image Studio v6.1. Threshold analysis enables identification of tissue in foreground and autofluorescence for background correction.

THE FUTURE OF DRUG DISCOVERY

The integration of organoids and spheroids into more drug discovery pipelines marks a significant advancement in

modern preclinical research. These 3D models offer several advantages over traditional 2D cultures, including more accurate prediction of drug responses and better representation of human tissue physiology. This shift in mindset and protocol has the potential to save billions of dollars wasted in clinical drug development, ultimately ensuring safer, more effective drugs get to patients faster. Given the FDA's recent guidance and the increased scrutiny on academic research and pharmaceutical regulatory frameworks, these innovative technologies are poised to play a pivotal role in advancing drug discovery, kickstarting precision medicine, and improving patient outcomes both now and in the near future.

This application note demonstrated capabilities of the brand-new Atlas™ Imaging System. This imager delivers both higher-throughput screening and higher-resolution 5 µm multiplex imaging for a fraction of the cost of a confocal microscope or flow cytometer.

The examples highlighted above showcase the Atlas Imager's versatility across diverse research applications—from cell viability studies and spheroid imaging to multiplexed fluorescence assays.

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LICORbio ATLAS™ IMAGER

Data-rich 2D and 3D Cell-Based Imaging, Faster Than Ever.